CSCI-6730 OPERATING SYSTEMS

Project #2 – Concurrency Bug Investigation

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INTRODUCTION

In this assignment, we try to find potential concurrency bugs such as race condition, deadlock and livelock and try to find a solution to solve this bug in four different applications namely – worker, stringbuffer, pbzip2 and aget. We further try to investigate, reproduce and resolve this bug in these applications.

PROGRAM – 1: WORKER

1. **Problem Identification:**

   In this program, if a thread accesses memory after it has been freed by the signal handler, it can cause a segmentation fault, which is a race condition involving memory access after deallocation.

2. **Bug Reproduction**:

   To trigger a segmentation fault in this multithreaded program, we introduce sleep() in order to disturb the normal synchronization and timing among threads and memory access.

   Inside the worker thread, we placed sleep(1) call before accessing thread_data[tid].data, which introduces a race condition between the threads. Since we introduced a delay for access to thread_data, signalHandler might free the memory when another thread tries to access it, which would create a segmentation fault.

```
    sleep(1);

    thread_data[tid].data[0]
    += rand()%10;
    thread_data[tid].data[1]
    += rand()%10;
    thread_data[tid].data[2]
    += rand()%10;
    usleep(1000);
```

This happens if we press Ctrl+C, the signal handler will make thread_states = 0 and start freeing the memory. Since the worker thread is still accessing the thread_data, whereas signalHandler is freeing it, we will get a segmentation fault because of accessing the already freed memory.

```
[hm92170@csci-odin worker]$ make
cc -g -lpthread  -o worker worker.c
[hm92170@csci-odin worker]$ ./worker
^CMain: caught SIGINT (Ctrl+C). Exiting gracefully.
Segmentation fault (core dumped)
```

3. Root cause analysis

From the bug report, it is observable that a race condition is occurring when each of several worker threads in worker function tries to access and update the thread_data[tid].data simultaneously. Each of the worker threads modifies the same data[0], data[1] and data[2] without any locking or protection mechanism like mutex. It may lead threads to overwrite the updates of other threads and introduce inconsistencies in the thread_data.

If Ctrl+C is pressed, a segmentation fault will occur because the signal handler frees memory while worker threads are still accessing the thread_data[tid].data. This is due to worker threads dereferencing memory that has already been freed. Threads might try to access memory that has been freed.

4. Fix Design

For resolving the segmentation fault, mutex locks have been implemented to sustain thread-safe, that is, non-interfering operations on shared data. Mutex lock effectively addresses the concurrency issues of the program.

a. Mutex initialization and destruction

We initialize mutex before its use and destroyed after all the threads are finished, it is destroyed to avoid undefined behaviour. Here, critical section is reading and writing into shared data structures such as thread_data and total variable.

By defining the mutex globally, all threads share the same lock. A call to pthread_mutex_init() is used to initialize the mutex before any thread starts working with it.

```
pthread_mutex_init(&lock, NULL); // Initialize the
mutex

for (i = 0; i < MAX_THREAD; i++) {
    // Create threads
    pthread_create(&thread[i], NULL, worker,
    (void*)i);
}

for (i = 0; i < MAX_THREAD; i++) {
    pthread_join(thread[i], NULL);
    usleep(100000);
}

printf("Total = %d\n", total);
pthread_mutex_destroy(&lock); // Destroy the mutex
return 0;
```

b. Mutex locking in signalHandler

We use locks here while updating shared data total, which prevents race conditions. One thread might be reading and updating the total, while at the same time another is doing the same. And also, without mutex lock, a thread could try to update thread_data at the same time when signalHandler is freeing that memory. This may bring about segmentation faults.

```
pthread_mutex_lock(&lock);
for (i = 0; i < MAX_THREAD; i++) {
    total += thread_data[i].data[0];
    total += thread_data[i].data[1];
    total += thread_data[i].data[2];
    free(thread_data[i].data);
}
free(thread_data); // Free the thread_data array
pthread_mutex_unlock(&lock); // Unlock the mutex
```

c. Mutex in worker threads:

In the worker function, shared data - thread_data[tid].data is modified within the critical section. The mutex is locked before modification and unlocked afterwards. This way several threads will not be able to concurrently write to the same memory location, hence solving the segmentation fault.

```
// Lock the mutex before updating shared data
pthread_mutex_lock(&lock);
thread_data[tid].data[0] += rand() % 10;
thread_data[tid].data[1] += rand() % 10;
thread_data[tid].data[2] += rand() % 10;
pthread_mutex_unlock(&lock); // Unlock the
mutex

usleep(1000); // Sleep for 1 millisecond
```

5. **Discussion of Fixes:**

The solution to this problem is the utilization of mutex locks, which effectively solves the segmentation fault and the concurrency problems in this multi-threaded program. Here's why it works:

a. Prevents race condition: Mutex locks ensure that at any given time, only one thread can be inside to access critical section updating total or thread_data. This prevents multiple threads from concurrently modifying shared data, which may lead to inconsistent states and segmentation faults.

b. Synchronization: multiple threads share resources, so a mutex must be used to protect the access to total and thread_data.

c. Safe signal handling: the signalHandler cleans up shared resources, thread_data and total, using the mutex locks while handling the termination signal, SIGINT.

Potential for new bugs:

a. Deadlock might be caused by improper mutex locking - for example, when threads wait indefinitely for a lock that is never available. The unjustified synchronization causes the bugs to be very subtle and hard to debug.

b. Mutex operations (locking and unlocking) incur overhead, which can impact program performance if critical sections are accessed frequently.

c. When the number of threads increases, the management of mutes becomes more complex.

6. **Testing and Validation**

After applying mutex locks:

```
[hm92170@csci-odin worker]$ ./worker
^CMain: caught SIGINT (Ctrl+C). Exiting gracefully.
Thread #3: terminating..
Thread #1: terminating..
Thread #4: terminating..
Thread #7: terminating..
Thread #9: terminating..
Thread #8: terminating..
Thread #6: terminating..
Thread #2: terminating..
Thread #0: terminating..
Thread #5: terminating..
Total = 112837
[hm92170@csci-odin worker]$
```

PROGRAM – 2: STRINGBUFFER

1. **Problem Identification**:

In StringBuffer class, race condition is happening due to accessing/modifying of the shared variable value and count by multiple threads concurrently .

2. **Bug Reproduction:**

The code provided to us already raises the bug and gives assertion failed error. This happens when multiple threads access the same shared buffer.

3. **Root cause analysis:**

The following section clearly explains the identified bugs that are present in code and also the type of concurrency issues:

a. Destructor ~StringBuffer - The destructor deletes the value array without acquiring mutex lock. This can lead to undefined behavior or race condition where one thread is editing the buffer while the other one is deleting it.

b. Shared null_buffer - All threads share this static null_buffer, but no mutex was implemented to serialize access in case multiple threads append NULL. This can lead to an inconsistent state of the buffers.

c. Race condition in expandCapacity – Though expandCapacity is protected by mutex locks, the check of the append function whether newcount > value_length may cause a data race.

4. Fix Design:

a. Error handling in getChars, append and erase functions:

In the code provided, we used assert(0) for the handling of invalid input parameters. The program would just terminate immediately when it encountered any invalid conditions. In order to fix it, we updated conditions in the getChars() method to return false instead of asserting in the case of invalid indices.

```
if (srcBegin < 0 || srcEnd < 0 ||
srcEnd > count || srcBegin > srcEnd) {
    pthread_mutex_unlock(&mutex_lock);
    throw std::runtime_error("Invalid
    indices for getChars.");

}
```

b. Mutex locks in expandCapacity and print

One of the key operations was expanding the capacity of StringBuffer, which modified both the value and value_length attributes. We locked this method for protection from interruption by other threads, also trying to execute append or erase concurrently. Print is reading the internal state of StringBuffer; therefore, a lock should be placed to avoid inconsistent states.

```
void StringBuffer::expandCapacity(int minimumCapacity) {
    pthread_mutex_lock(&mutex_lock);
    int newCapacity = (value_length + 1) * 2;
    if (newCapacity < 0) {
        newCapacity = INTEGER_MAX_VALUE;
    } else if (minimumCapacity > newCapacity) {
        newCapacity = minimumCapacity;
    }

    char *newValue = new char[newCapacity];

    memcpy(newValue, value, count);
    delete[] value;
    value = newValue;
    value_length = newCapacity;
    pthread_mutex_unlock(&mutex_lock);
}
```

c. Error checking in append:

The code did not check for negative lengths or invalid inputs. Now we've added checks for negative lengths and throws std::runtime_error in case of invalid length.

```
int len = sb->length();
if (len < 0) { // Check for negative
length
    pthread_mutex_unlock(&mutex_lock);
    throw std::runtime_error("Invalid
    length for append.");
}
```

d. Thread safety in destructor:

The destructor did not use mutex lock for deleting value. In order to ensure that one thread does not try to access StringBuffer when another thread is deleting it, we add locks in destructor for deleting value.

```
StringBuffer::~StringBuffer() {

    delete[] value;
    pthread_mutex_destroy(&mutex_lock);
}
```

5. **Discussion of fixes:**

By doing the above fixes to the code, we try to address several critical issues associated with concurrency and memory safety.

a. Thread safety - Ensuring that access to shared resources in all places is properly synchronized with mutex locks reduces race conditions and corruption to a great extent.

b. Error handling – We return error states and throw exceptions instead of using assertions to make the code more reliable.

c. Memory management – Careful thought is given to the allocation and deallocation of memory.

Potential for new bugs:

a. Since multiple operations are attempting to update the shared value and count attribute without proper synchronization, race condition arises.

b. Shared variables - Both the value and count variables are shared by multiple threads but not all operations modifying them are guarded by mutex-locks. If at least two threads call methods like append or erase or print concurrently they interfere with each other.

c. if one thread calls expandCapacity while another thread is doing an append or erase the internal state of StringBuffer can become inconsistent.

d. In the append method, if one thread was appending data and another thread was checking on the length, the count could change unexpectedly during a read and write operation.

6. Testing and Validation

After appropriate fixes, we do not get the assertion error anymore, and the threads are correctly synchronized.

```
[hm92170@csci-odin stringbuffer]$ ./test
0th try - Thread 1: erase and append: 0
0th try - Thread 2: create and append: 0
100th try - Thread 1: erase and append: 1
100th try - Thread 2: create and append: 1
200th try - Thread 1: erase and append: 2
200th try - Thread 2: create and append: 2
300th try - Thread 1: erase and append: 3
300th try - Thread 2: create and append: 3
400th try - Thread 1: erase and append: 4
400th try - Thread 2: create and append: 4
500th try - Thread 1: erase and append: 5
500th try - Thread 2: create and append: 5
600th try - Thread 1: erase and append: 6
600th try - Thread 2: create and append: 6
700th try - Thread 1: erase and append: 7
700th try - Thread 2: create and append: 7
800th try - Thread 1: erase and append: 8
800th try - Thread 2: create and append: 8
900th try - Thread 2: create and append: 9
900th try - Thread 1: erase and append: 9
1000th try - Thread 2: create and append: 10
1000th try - Thread 1: erase and append: 10
1100th try - Thread 1: erase and append: 11
1100th try - Thread 2: create and append: 11
1200th try - Thread 1: erase and append: 12
1200th try - Thread 2: create and append: 12
1300th try - Thread 1: erase and append: 13
1300th try - Thread 2: create and append: 13
1400th try - Thread 1: erase and append: 14
1400th try - Thread 2: create and append: 14
1500th try - Thread 1: erase and append: 15
1500th try - Thread 2: create and append: 15
```

PROGRAM – 3: PBZIP2

1. Problem Identification and Root Cause Analysis:

The segmentation fault occurs occasionally during the execution of Pbzip2, primarily toward the end of execution.

Pbzip2 frequently locks and unlocks the MemMutex and OutMutex throughout the code. Possible race condition could take place while unlocking these mutexes without being properly locked in advance or having threads trying to unlock the same mutex simultaneously.

Another reason for the segmentation fault is due to the improper timing of destroying mutexes. The code calls pthread_mutex_destroy on MemMutex and OutMutex, but it's possible other threads could be using these mutexes at this point.

There are various cases where memory is being freed by the function delete, such as FileData, OutputBuffer. If memory is deallocated while other threads are still accessing or using this data, a race condition is occurring.

2. **Bug reproduction**:

   To reproduce the issue, we inserted sleep() calls in both the consumer function and the main function to increase the likelihood of the race condition and, consequently, the segmentation fault.

   a. In consumer function, we added sleep(1) after setting the variable allDone to 1. This will introduce some delay before unlocking the mutex and thread termination in the consumer thread to provide room for a race condition during the thread termination.

```
                    if (allDone == 1)
                    {
                        sleep(1);
                        pthread_mutex_unlock(fifo->mut);
#ifdef PBZIP_DEBUG
                        printf ("consumer: exiting2\n");
#endif
                        return (NULL);
                    }
```

   b. In main function, just before reclaiming the memory and destroying mutexes (OutMutex and MemMutex), sleep(1) call was inserted.

```
    // reclaim memory
    queueDelete(fifo);
    fifo = NULL;
    if (OutMutex != NULL)
    {
            pthread_mutex_destroy
            (OutMutex);
            delete OutMutex;
            OutMutex = NULL;
    }
    if (MemMutex != NULL)
    {
            pthread_mutex_destroy
            (MemMutex);
            delete MemMutex;
            MemMutex = NULL;
    }
sleep(1);
```

This artificial delay triggers the segmentation fault by creating a situation where the threads' access patterns collide with the cleanup process, making the underlying race condition easier to observe.

```
[hm92170@csci-odin pbzip2]$ ./pbzip2 -k -f test.file
Parallel BZIP2 v0.9.4 - by: Jeff Gilchrist [http://compression.ca]
[Aug. 30, 2005]             (uses libbzip2  by Julian Seward)

* This is a BETA version - Use at your own risk! *

         # CPUs: 36
 BWT Block Size: 900k
File Block Size: 900k
-------------------------------------------
         File #: 1 of 1
     Input Name: test.file
    Output Name: test.file.bz2

     Input Size: 2930969 bytes
Compressing data...
    Output Size: 875176 bytes
-------------------------------------------
Segmentation fault (core dumped)
[hm92170@csci-odin pbzip2]$
```

PROGRAM – 4: AGET

1. Problem Identification:

The atomicity violation in Aget's download code arises because multiple threads have accessed the shared resources such as a variable 'bwritten' non-atomically. In the present case of code, atomicity violations arise because of the interplay of shared resource handling improperly, log update interruptions, and usage of asynchronously canceling the threads.

2. Bug Reproduction:

To consistently reproduce the atomicity violation bug and thus trigger the log corruption issue, we have injected a deliberate delay into the code by inserting a sleep(1) before updating the bwritten variable: As this increases the probability of interruptions during critical instants of downloading, when stopping the download and then resuming it later, this would result in a corrupted log file.

```
td->offset += dw;
sleep(1);
pthread_mutex_lock(&
bwritten_mutex);
bwritten += dw;
pthread_mutex_unlock(&
bwritten_mutex);
```

The purpose of this modification is to introduce a delay, giving enough time for an external interruption (e.g., ctrl+c) to occur during a critical section where bwritten is being updated. This makes it more likely to stop the download process while the variable is in a partially updated state.

```
[hm92170@csci-odin aget]$ ./aget -n2 http://ftp.us.debian.org/debian-cdimage/12.7.0/i386/iso-cd/debian-edu-12.7.0-i386-netinst.iso -l ./test.download
Aget > Attempting to read log file for resuming download job...
Aget > Couldn't find log file for this download, starting a clean job...
Aget > Head-Request Connection established
Aget > Downloading /debian-cdimage/12.7.0/i386/iso-cd/debian-edu-12.7.0-i386-netinst.iso (683671552 bytes) from site ftp.us.debian.org(64.50.233.100:80).
Number of Threads: 2
File already exists! Overwrite?(y/n) y
^C^C caught, terminating download job. Please wait...00% done] Bytes: 28562
Download job terminated. Now saving download job...
--> Logfile is: /home/myid/hm92170/test.download-ageth.log, so far 44946 bytes have been transferred
[hm92170@csci-odin aget]$ ./verifier /home/myid/hm92170/test.download-ageth.log
/home/myid/hm92170/test.download-ageth.log is incorrect and may be corrupted. Deleting the log.
Total bytes written: 44946, Size of content: 61330 bytes.
[hm92170@csci-odin aget]$
```

3. **Root cause analysis:**

   a. Shared variable bwritten: This variable is shared between multiple threads for keeping track of the total bytes written. There is a mutex associated with it known as bwritten_mutex to guard against concurrent accesses. If there is an interruption in download between the lock and unlock operation, bwritten may not hold the actual number of bytes. This will result in inconsistency in the log file.

   b. Signal Handling: The function handle_signal(int signal) captures the SIGINT signals that are caused, for example, when the user stops the download with ctrl+c. However, it is not ensured that the currently executed operations - for example, the increase of bwritten - will be finished before the program will be stopped. This can lead to a partially updated value of bwritten, thus the operation is not atomic.

   c. Potential race condition: Because downloading and writing data involve multiple threads, the race condition can be easily welcomed when two or more threads make an attempt to update the bwritten/log file without having proper synchronization. It uses mutexes but does not use them on all critical sections, such as handling thread cancellations and signals.